

QUESTION PAPER

Assessment Name : Solar Panel Installation Technician Theory Set 2

Duration :

Total Marks : 160

Q1 .What should Hemant ensure after installing a junction box?

- 1) That it is covered 2) That it is getting sufficient sunlight 3) That it is getting at least the minimum amount of sunlight 4) That it is exposed for ensuring cooler temperatures

Q2 .Ram is installing a solar PV system in Mumbai. He wants maximum efficiency. Therefore, he should ensure that the solar panels are facing _____ direction.

- 1) East 2) South 3) North-East 4) North-West

Q3 .Rakesh is carrying a solar cell of 15 percent efficiency with an area of 1 meter square. How much power should Rakesh expect from the cell if the angle of incidence is 60 degrees to the normal?

- 1) 75 W 2) 150 W 3) 100 W 4) 200 W

Q4 .Which of the following factors will not affect the different types of mountings?

- 1) Load generated 2) Area covered 3) Cost of the system 4) Load requirement

Q5 .Mayank is troubled by a personal issue. He is not able to properly concentrate on his work. What should he do if he wants to leave?

- 1) Inform the supervisor, take permission and then leave 2) Leave straightaway 3) Inform the supervisor when he has left 4) Tell the customer and leave

Q6 .Which of the following steps comes earlier than the others when installing a flat surface mounting?

- 1) Levelling the land 2) Constructing the mounting 3) Installing PV modules 4) Measuring the PV system output

Q7 .What is the benefit of using solar tracker technology over fixed tilt mounting?

- 1) Enhances light capture 2) Decrease the chances of short circuit 3) Increases the strength of mounting 4) Cost effective

Q8 ."Amit has decided that the mounting constructed should be effective throughout the day and be least affected by the movement of the sun. Which type of mounting should he construct?"

- 1) Solar panel-fixed mounting 2) Fixed solar mounting facing west 3) Fixed solar mounting facing east 4) Solar tracker mounting

Q9 .Vikrant is climbing up the ladder to reach the roof. How many points of contact with the ladder will ensure his safety at all times?

1) 4 2) 2 3) 1 4) 3

Q10 .If Ajit's customer wants solar panels with the highest efficiency, which of the following solar panels should he collect?

1) Mono crystalline 2) Poly crystalline 3) Thin film 4) Amorphous crystalline

Q11 .Following safety regulations is important for both employees and the company.

1) YES 2) NO

Q12 .Ramesh is selecting the frame for mounting. Which of the following factors is he least likely to consider during the process?

1) Output of PV system 2) Size of PV modules 3) Type of mounting 4) Price of frame

Q13 .Yuvraj is a solar panel installation technician. He is told by his manager that low quality results will be accepted only if there are some constraints.

1) TRUE 2) FALSE

Q14 .How should Kishore set the multimeter if he wants to check the current in the solar system?

1) Set it to 'watts' and touch the multi-meter's red lead to positive terminal and black to negative 2) Set it to 'Ampere' and touch the multimeter's red lead to positive terminal and black to negative 3) Set it to 'watts' and touch the multimeter's black lead to positive terminal and red to negative 4) Set it to 'current' and touch both the multimeter's leads to the positive terminal

Q15 .What must Uday ensure about parts of modules when installing solar panels?

1) They should be disassembled at least once 2) They should be assembled at least once 3) They should not be assembled or disassembled 4) They must be simultaneously disassembled and assembled

Q16 .All of the following characteristics or behavioural qualities can help someone excel at the workplace except _____.

1) Being polite 2) Good stance or body language 3) Being honest 4) Being impulsive

Q17 .Ravish is installing a solar PV system in Delhi, which is at latitude 28. What tilt angle should he ensure during winters and which is the best time to adjust the winter angle?

1) Angle 43 degrees; adjustment should be in mid-September 2) Angle 43 degrees; adjustment should be in the beginning of October 3) Angle 13 degrees; adjustment should be in mid-September 4) Angle 13 degrees; adjustment should be in the beginning of October

Q18 .What kind of a tracking system should be used to ensure that sunlight falls perpendicular to the PV module throughout the year?

1) Single axis tracker 2) Two axis tracker 3) Three axis tracker 4) Four axis tracker

Q19 .While handling the electrical components of a solar PV system, Neeraj should avoid using which of the following PPEs?

- 1) Rubber shoes 2) Metal gloves 3) Rubber gloves 4) Protective clothing

Q20 .For fixing the mount, Firoz wants to use a device that can bind and press panels and mounting rails together. Which of the following devices can he use to complete his task?

- 1) Welding machine 2) Tracker 3) Lever 4) Clamp

Q21 .Only physical damage to solar panels can affect their electricity generation capability.

- 1) TRUE 2) FALSE

Q22 .Anil gets a 0.5 percent incentive on the total cost of solar PV systems he has installed in a month. If he gets an incentive of Rs. 10,000 for a certain month, what was the total cost of the systems he has installed?

- 1) Rs. 10 Lakhs 2) Rs. 20 Lakhs 3) Rs. 15 Lakhs 4) Rs. 25 Lakhs

Q23 .Which of the following is the main issue associated with installation of a solar PV system?

- 1) Unavailability of solar panels 2) Unavailability of DC connecting wires 3) High installation and material cost 4) Difficulty in finding unshaded areas

Q24 .Sohan is following the 7-step installation process for installing a solar PV system. If each step takes 45 minutes of his time, how much time will he require to complete the installation work?

- 1) 5 hours 2) 5 hours 15 minutes 3) 5 hours 30 minutes 4) 6 hours

Q25 .Ravi is showing a single solar cell to the customer. He tells him that the voltage of the single solar cell is _____.

- 1) 1 V 2) 1.5 V 3) 0.5 V 4) .8 V

Q26 .Ramesh is planning his week's work. What is the first thing he should do?

- 1) Determine his strengths and weaknesses 2) Evaluate alternatives 3) Set an objective 4) Seek feedback

Q27 .Sanant has to pick up a heavy box from the floor. What should be the ideal position of his back?

- 1) Upright 2) Twisted sideways 3) 45 degrees to the floor 4) Tilted towards the load

Q28 .Hemant has to install a Field Junction Controller in a solar PV system. What is the main use of such a device?

- 1) It maximises the light captured by solar cells 2) It acts a interface between the solar panels and charge controller 3) It is useful for connecting cells in a series arrangement 4) It is useful in connecting cells in a parallel arrangement

Q29 .Surya has a discussion with Logistic Department. He realises that holding safety stocks as part of the inventory is essential. What is the most likely reason for holding such stocks?

- 1) For catering to demand-forecast errors 2) For catering to a large storage capacity 3) To improve stock-accuracy levels 4) To improve stock-holding levels

Q30 .Ishant finds that a solar panel is not working properly due to damage caused during the removal of packaging material. Which of the following factors is least likely to get affected by this issue?

- 1) Load generated by panels 2) Warranty on panels 3) Load requirement 4) Working of panels

Q31 .Why must you be punctual every time when meeting clients or boss?

- 1) To show them that you are good at time management 2) To show them that you miss them 3) To show them that you are serious about your work 4) To show them that you don't want to waste time

Q32 .Use of all of the following materials can ensure safety except _____.

- 1) Monitoring equipment 2) Welding equipment 3) PPE 4) Insulated gloves

Q33 .If Hari is reporting problems that he has faced in the field, which of the following is he least likely to report?

- 1) Insufficient output 2) Shaded areas 3) Malfunctioning of equipment 4) Delay in salary

Q34 .Yagya tries to understand the customer's preferences when interacting with customers. What should he do for the same?

- 1) Ask for output requirement 2) Ask relevant questions 3) Seek solar PV system's output 4) Refer MSDS

Q35 .Vinod is a solar PV installation technician. He need not learn which of the following tasks.

- 1) Undertaking first-aid 2) Giving CPR 3) Construction of mounting 4) Baking of bricks

Q36 .After receiving solar panels from the vendor, Ishant wraps them in an opaque sheet. What is the main reason behind such covering?

- 1) To avoid accumulation of dust particles 2) To avoid breakage 3) To ensure that the panel is not affected by water or humidity 4) To ensure that there is no electricity generation before installation

Q37 .BOS (Balance-Of-System) is defined as everything except _____.

- 1) Module support structure 2) Energy storage batteries 3) PV modules and the load 4) Power conditioning equipment

Q38 .Karn is expected to install a solar PV system. If the customer wants ground mounting, which of the following factors should Ravi ensure?

- 1) Large garden or field 2) Firm roof 3) Soil fertility 4) Junction box is installed in a tree

Q39 .Naman can avoid all of the following hazards except _____ for ensuring zero accidents at the workplace.

- 1) Environmental hazards 2) Physical hazards 3) Electrical hazards 4) Accidents

Q40 .Before working on a solar module, Hitesh should ensure that the current flowing in the module is _____.

- 1) At least 1 A 2) 2 A 3) 0 A 4) It doesn't matter

Q41 .Karan is likely to come into contact with a highly hazardous material when handling which of the following solar PV components?

- 1) Solar tracker 2) PPE 3) Batteries 4) Mounting tools

Q42 .For fixing the mount, Firoz wants to use a device that can bind and press panels and mounting rails together. Which of the following devices can he use to complete his task?

- 1) Welding machine 2) Tracker 3) Lever 4) Clamp

Q43 .Sanjay knows that inverter rating should be 25 percent more than the load of the house / building if he recommends an inverter of rating 1000 W to the customer. What is the total load such an inverter is expected to handle?

- 1) 1000 W 2) 1200 W 3) 750 W 4) 800 W

Q44 .Vikrant's supervisor tells him that as a solar panel installation technician, he has to take part in all of the following processes except _____.

- 1) Solar panel manufacturing process 2) Solar panel installation process 3) Solar panel maintenance process 4) Solar panel testing process

Q45 .What should Ravi look for if he wants detailed instructions on a particular equipment used in a solar PV system?

- 1) Life saving appliances booklet 2) Log sheet 3) Personal book of supervisor 4) User's manual for equipment

Q46 .Surya takes four hours to install a single solar PV system. He is required to install 30 systems in a month. If he works on an 8-hour shift, how many days will he take to achieve his monthly target?

- 1) 15 days 2) 14 days 3) 16 days 4) 12 days

Q47 .If Karan has to complete a hazardous task, which of the following documents must he complete first before carrying on with his work?

- 1) System reading log 2) Safety checklist 3) Overtime log 4) Time sheet

Q48 .Pranay has been advised to use large wire sizes in battery interconnections. Why is Pranay still using them when he knows that they are a little bit oversized?

- 1) Because they provide a lower resistance path for battery short circuit currents 2) Because they allow for increases in load size or array size 3) Because they keep all battery currents as equal as possible 4) Because they secure the batteries in case of high winds

Q49 .Ram has completed an electrical installation, which includes a PV power supply system without simple separation between the AC side and DC side. What should he install to provide fault protection by automatic disconnection of supply?

- 1) Reverse Current Device (RCD) 2) Residual Current Device (RCD) 3) Reserve Current Device (RCD) 4) Removable Current Device (RCD)

Q50 .Rachit works on an 8-hours-a-day shift and can install a small solar PV system in 6 hours. How many systems can he install in a week with an off on a Sunday?

- 1) 6 2) 8 3) 10 4) 7

Q51 .Which of the following practices could lead to a fire hazard?

- 1) Welding near non-flammable substances 2) Welding anywhere 3) Welding near inflammable substances 4) Welding near rocks

Q52 .All of the following statements are true about ESD except that _____.

- 1) It is sudden flow of electricity between two electrically charged objects 2) It cannot be caused by contact 3) It can be caused by contact 4) It can be caused by electrical short

Q53 .Which of the following statements is true about an azimuth angle?

- 1) It is the horizontal angle measured clockwise from a south base line 2) It is the horizontal angle measured clockwise from a north base line or meridian 3) It is the horizontal angle measured anti-clockwise from a north base line or meridian 4) It is the vertical angle measured clockwise from a north base line or meridian

Q54 .Which of the following statements is true about tilt angles?

- 1) It is same throughout India 2) In Delhi, the tilt angle is greater than that of Mumbai 3) In Delhi, the tilt angle is lower than Mumbai 4) Power generation is not affected by tilt angle

Q55 .Varun finds that the customer's budget for solar PV installation is Rs. 4 Lakhs. However, if the cost of the system is Rs. 5 Lakhs, what is percentage increase in budget that Varun should seek from the customer?

- 1) 20 percent 2) 10 percent 3) 30 percent 4) 25 percent

Q56 .Amar has some issues with the training methods. If he wants to resolve those issues, he must discuss the same with which of the following departments?

- 1) Accounts 2) Admin 3) HR 4) Logistics